

# GANESH GUPTA

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## Experience

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### Oracle

Jun 2025 – Present

*Software Engineer, AI Platform*

- Provisioned and operated a large-scale multi-cloud Linux fleet (AWS, GCP, CoreWeave) supporting production training and inference workloads, keeping systems healthy and scalable through automation and self-service tooling.
- Wrote operational tooling and automation in Go and Python to reduce manual intervention: scheduling, resource allocation, and fault-tolerant placement logic across the fleet, version-controlled with Git.
- Built observability and autoscaling for fleet health using Prometheus and Grafana, debugging complex systems issues and driving them to resolution to keep utilization and latency SLOs stable through node loss.

### QureAI

Mar 2026 – Present

*ML Infrastructure Engineer*

- Cut p99 LLM inference latency 420ms to 140ms across 50K+ daily requests by profiling and migrating synchronous PyTorch-backed serving to async vLLM, optimizing GPU throughput under autoscaling.
- Built a model performance and observability framework tracking latency, drift, and evaluation regressions across production model versions.

### Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

*Software Engineer*

- Built and operated 25+ production C++ and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.
- Standardized failure-handling contracts across 25+ services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

## Projects

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### Distributed KV Store / VPC Control Plane Simulator | [github.com/ganeshhgupta/distributed-kv-store](https://github.com/ganeshhgupta/distributed-kv-store)

- Built a Raft-consensus key-value store in-house with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

### Watch-your-LLM | [github.com/ganeshhgupta/watch-your-llm](https://github.com/ganeshhgupta/watch-your-llm)

- Production performance and eval tooling with rubric-based scoring, drift detection, and per-call latency benchmarking across model versions; FastAPI backend, PostgreSQL.

## Education

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### University of Texas at Arlington

Aug 2023 – May 2025

*MS, Computer Science, GPA 4.0/4.0*

### MCKV Institute of Engineering

Jul 2016 – Jul 2020

*BTech, Computer Science*

## Skills

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**Languages:** Python, Go, C++, Java, SQL

**Systems & Automation:** Linux Fleet Operations, Operational Tooling, Automation Scripting, Fault Tolerance, Autoscaling, Distributed Systems

**Observability & Ops:** Prometheus, Grafana, Git, CI/CD, Incident Debugging

**Cloud & Infra:** AWS (EC2, Lambda, S3), GCP (Cloud Run), Docker, Kubernetes

## Certifications

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AWS Certified Developer Associate | GCP Professional MLOps Engineer